Guru Naveen Reddy Siddu

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Verilog — SystemVerilog — UVM

Professional Summary

- Entry-level Hardware Engineer with hands-on experience across the **VLSI design flow – RTL design, digital logic implementation, and functional verification** – built through IP-level chip projects taken from specification to sign-off.
- Took a **PCIe Gen-2 SerDes** block from RTL through **100% functional coverage and verification sign-off** using a self-built UVM environment, demonstrating the full design-to-validation lifecycle.
- Strong foundation in **digital logic design**: combinational and sequential circuits, finite-state machines, multiplexers, and register-level design, applied across multiple RTL projects (SRAM, UART, AXI, MUX).
- Skilled in **functional verification and pre-silicon validation**: constrained-random verification, assertion-based verification (SVA), coverage-driven verification, regression execution, and waveform-based root-cause debugging using QuestaSim.

Technical Skills

Digital Logic Design: Combinational & Sequential Circuits, Finite State Machines, Multiplexers, Flip-Flops, Registers, Timing Fundamentals.

VLSI Design Flow: RTL Design → Functional Verification → Coverage Closure → Sign-off (hands-on); conceptual understanding of Synthesis and Physical Design stages.

HDL / HVL: Verilog, SystemVerilog, SystemVerilog Assertions (SVA).

Functional Verification & Validation: UVM, Constrained Random Verification, Assertion Based Verification (ABV), Coverage Driven Verification (CDV), Functional Coverage, Code Coverage, Coverage Closure, Regression Testing, Pre-Silicon Validation.

Programming & Scripting: Python (signal processing, data/statistical analysis).

Protocols / IPs: PCIe Gen-2 (LTSSM, 8b/10b Encoding, SerDes), APB, AXI, UART.

EDA Tools: QuestaSim, ModelSim.

Version Control & OS: Git, Linux

Projects

PCIe Gen-2 SerDes – Digital Design & Verification *Verilog — SystemVerilog — UVM — QuestaSim*

Designed and verified the PCIe Gen-2 Physical Layer SerDes – transmitter and receiver sub-blocks – with a reusable UVM environment built to validate protocol compliance and data integrity.

- Designed **Scrambler, 8b/10b Encoder, PISO, SIPO, 8b/10b Decoder,** and **Descrambler** digital sub-blocks in Verilog.
- Built a reusable **UVM environment** (sequencer, driver, monitor, scoreboard, agent, env, tests) from scratch for pre-silicon functional validation.
- Wrote **SVA** and functional coverage models to verify **LTSSM, 8b/10b encoding/decoding, TLPs, DLLPs,** and Configuration Space (Type-0/Type-1) behavior.
- Achieved **100% functional coverage** and closed verification sign-off through regression execution and waveform-based root-cause debugging in QuestaSim.

APB IP Core – Digital Design Verification *SystemVerilog — UVM — QuestaSim*

Verified the APB finite-state machine (Idle → Setup → Access) across single- and multi-slave communication, with focus on `PREADY`-driven wait-state handling.

- Architected a class-based UVM environment from a self-authored verification plan / feature extraction sheet.
- Verified write/read operations with and without wait states, same-slave and cross-slave communication, and **20 back-to-back** read/write transactions.
- Covered random, constant, increment, and decrement address/data patterns to stress-test protocol edge cases.
- Closed **functional and code coverage** for sign-off; wrote SVA for reset, read/write, and state-transition checks.

UART Transmitter & Receiver with FIFO – Digital Design *Verilog — SystemVerilog — QuestaSim*

Designed a full-duplex UART – transmitter and receiver – each paired with its own synchronous FIFO, built from fundamental digital logic (finite-state machines, shift registers, pointer-based buffering).

- Designed a **12-state transmitter FSM** with bit-serial shifting, XOR-based parity generation, and a ready/`flag` handshake for FIFO-driven streaming.
- Designed a **5-state receiver FSM** (Idle, Start, Data, Parity, Stop) with parity checking and stop-bit validation.
- Built **parameterized synchronous FIFOs** (pointer logic, occupancy counter, full/empty flags) for both TX and RX datapaths.
- Confirmed bit-accurate serial timing on both paths through simulation and waveform debugging in QuestaSim.

Certifications & Training

- **RTL Design & Verification Training Certificate** – VLSI FIRST Technology, Bengaluru
- Hands-on program covering the VLSI design flow – digital logic, RTL design, and functional verification – using Verilog, SystemVerilog, UVM, and SVA, applied directly to the PCIe Gen-2, APB, and UART projects above.

Publications

- **Cardiac Sound Enhancement using Adaptive Filtering and Wavelet Decomposition** – IEEE ICCSP 2024
  Python, Signal Processing
- Co-authored and documented a research project evaluating system performance using MSE, RMSE, and SNR metrics; prepared structured technical documentation and presented findings to an academic audience.

Education

B.Tech in Electronics & Communication Engineering **2021 – 2024**
Madanapalle Institute of Technology and Science, Andhra Pradesh *CGPA: 8.4 / 10.0*